

ALEKSANDRA KARPOVA

Software Engineer

Pavia, Italy [open to relocation]

• +39 351 436 42 79 • aleksandra.karpova042@gmail.com

• LinkedIn: [linkedin.com/in/alekskarpova](https://www.linkedin.com/in/alekskarpova)

Software Engineer specializing in real-time distributed systems and machine learning, with experience building production-grade data pipelines and deploying ML models in high-stakes industrial environments. Skilled in designing low-latency backend systems that process high-frequency sensor data and drive automated decision-making. Strong focus on system reliability, scalability, and clean architecture, with hands-on experience integrating ML into live control systems.

Work Experience

Software Engineer & ML Engineer– Industrial & Autonomous Systems

OPTECS B.V. | Rotterdam, Netherlands

February 2025 - Present

- Designed and implemented real-time backend systems processing high-frequency sensor data (sub-second to second-level updates) across multiple industrial data streams
- Built scalable time-series data pipelines supporting continuous monitoring and control of large-scale mining equipment, handling dozens of signals per system
- Deployed machine learning models into production for predictive monitoring and anomaly detection, contributing to earlier detection of system anomalies and reduced risk of unplanned downtime
- Developed a cross-platform OPC UA integration layer, enabling low-latency (<~100 ms) communication between heterogeneous industrial systems
- Refactored legacy components and increased test coverage from 50% to 85%, reducing regression issues and improving system stability in production
- Optimized system performance and code efficiency, achieving consistent low-latency execution suitable for real-time decision-making environments
- Improved system robustness by handling edge cases in streaming data (missing values, noise, outliers), ensuring stable operation under non-ideal conditions
- Collaborated in cross-functional teams to deliver production-grade systems operating under strict reliability and uptime requirements

Projects Experience

Industrial Mill Power Prediction System

- Developed and evaluated regression models (SGD, Ridge, Lasso, XGBoost) for predicting equipment power consumption
- Engineered features from time-series sensor data to improve model performance
- Built FastAPI-based inference service for real-time predictions in production-like environments

Satellite Telemetry Tracking & Control System

- Designed a system for real-time acquisition and processing of satellite telemetry data
- Implemented data pipelines for monitoring position, velocity, and system health parameters
- Developed backend services for telemetry visualization and control signal handling
- Established foundation for closed-loop control and anomaly detection in autonomous systems

Education History

October 2022 - July 2026

BSc in Artificial Intelligence

University of Pavia, University of Milan, University of Milan-Bicocca

- Thesis: Quantum Machine Learning and QJPEG for Image Classification
- Vice President, AIESEC International (2024–2025)
- Samsung Innovation Campus – AI & IoT (2023–2024)

Skills

Programming & Backend:

- Python (Advanced), Java, Dart (Flutter), C#, C++
- FastAPI, REST APIs, Django, Flask, Pydantic
- Distributed systems, real-time processing, system design
- Docker, Git, CI/CD, unit testing
- OPC UA (industrial communication protocol)

Machine Learning & Data:

- Scikit-learn, PyTorch, TensorFlow
- NumPy, Pandas, SQL
- Time-series data processing, feature engineering, model deployment

Quantum Information:

- Qiskit, PennyLane

Certifications & Memberships

- Qiskit advocate – IBM Quantum
- AI-SEQ program
- IBM Machine Learning Professional

February 2026

September - October 2025

May 2024